



L2

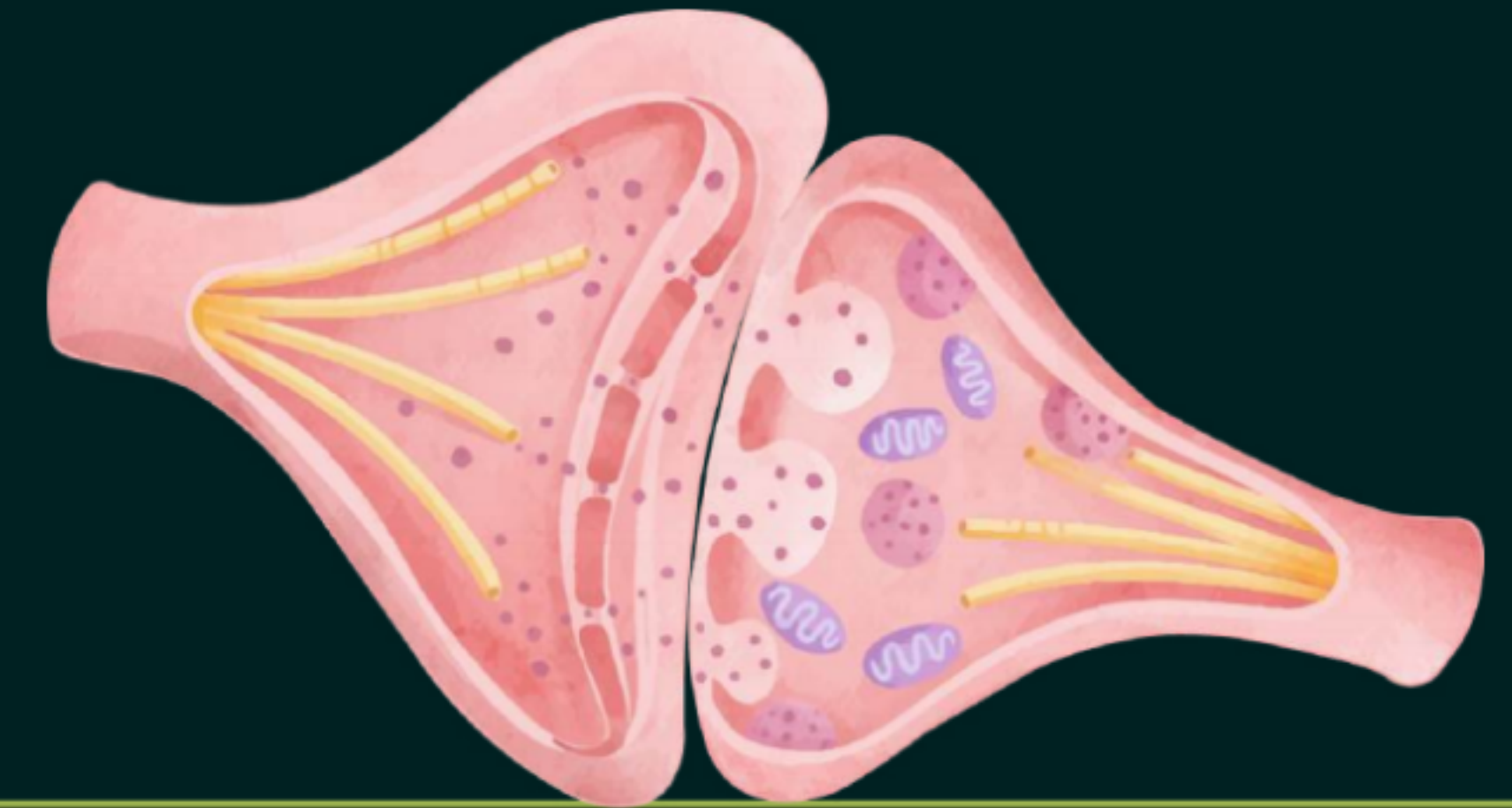
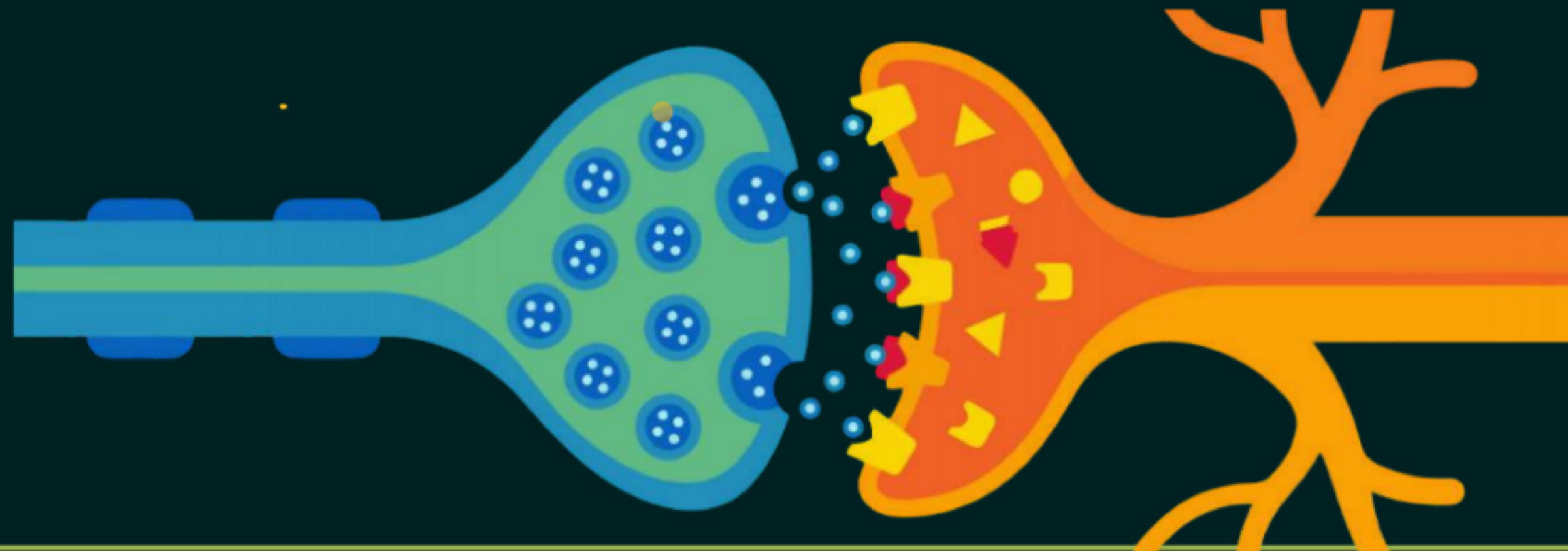
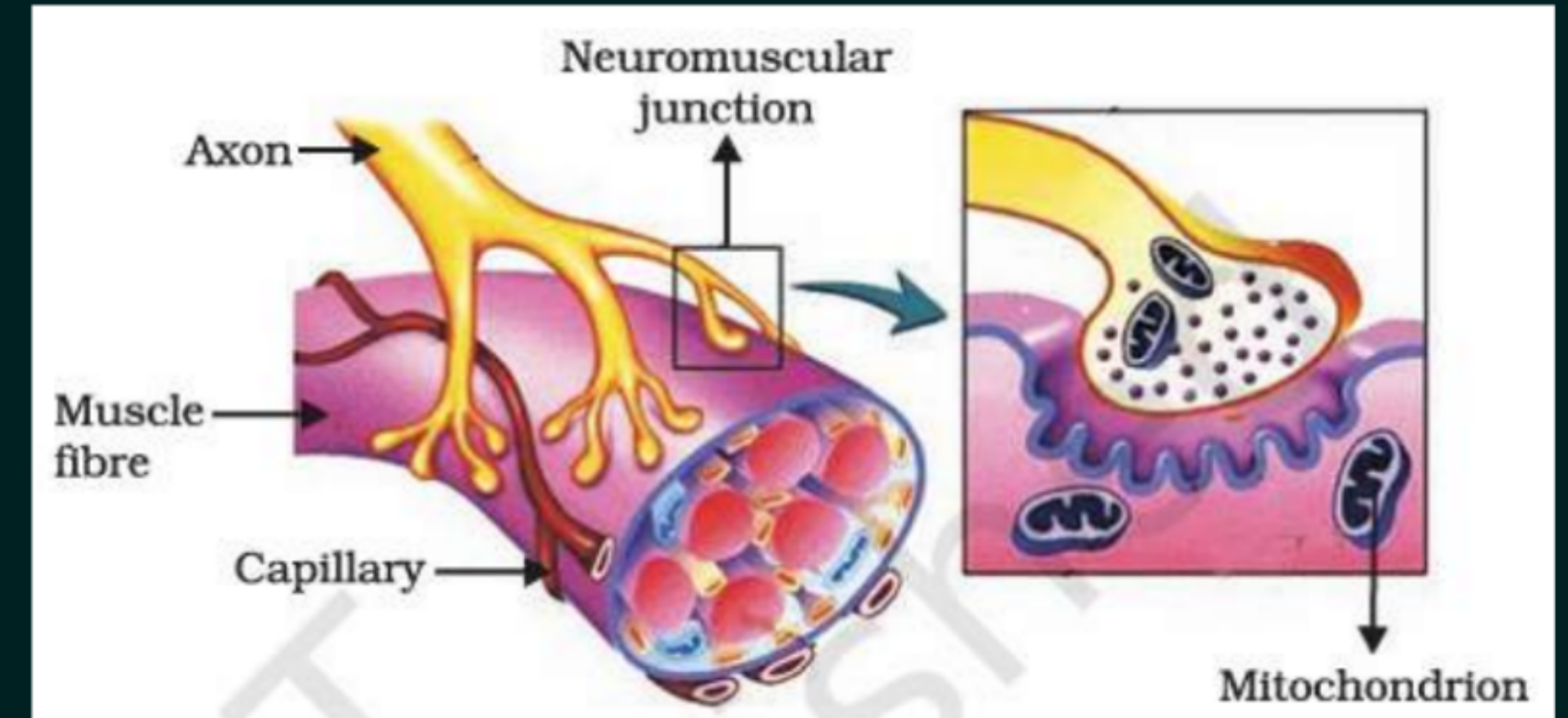
CLASS X - SCIENCE

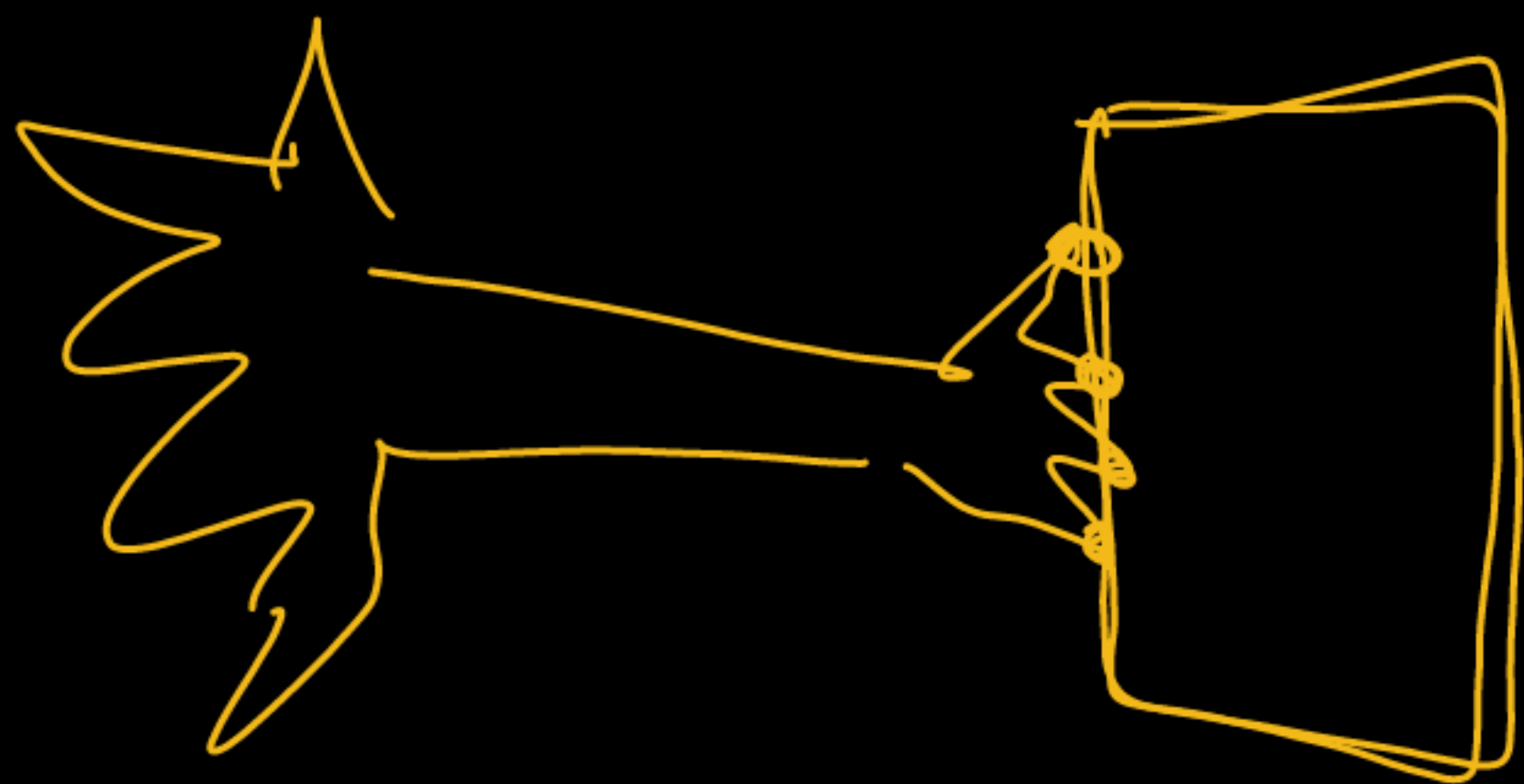
CONTROL AND COORDINATION

| PRASHANT KIRAD |

Neuromuscular Junctions (NMJ)

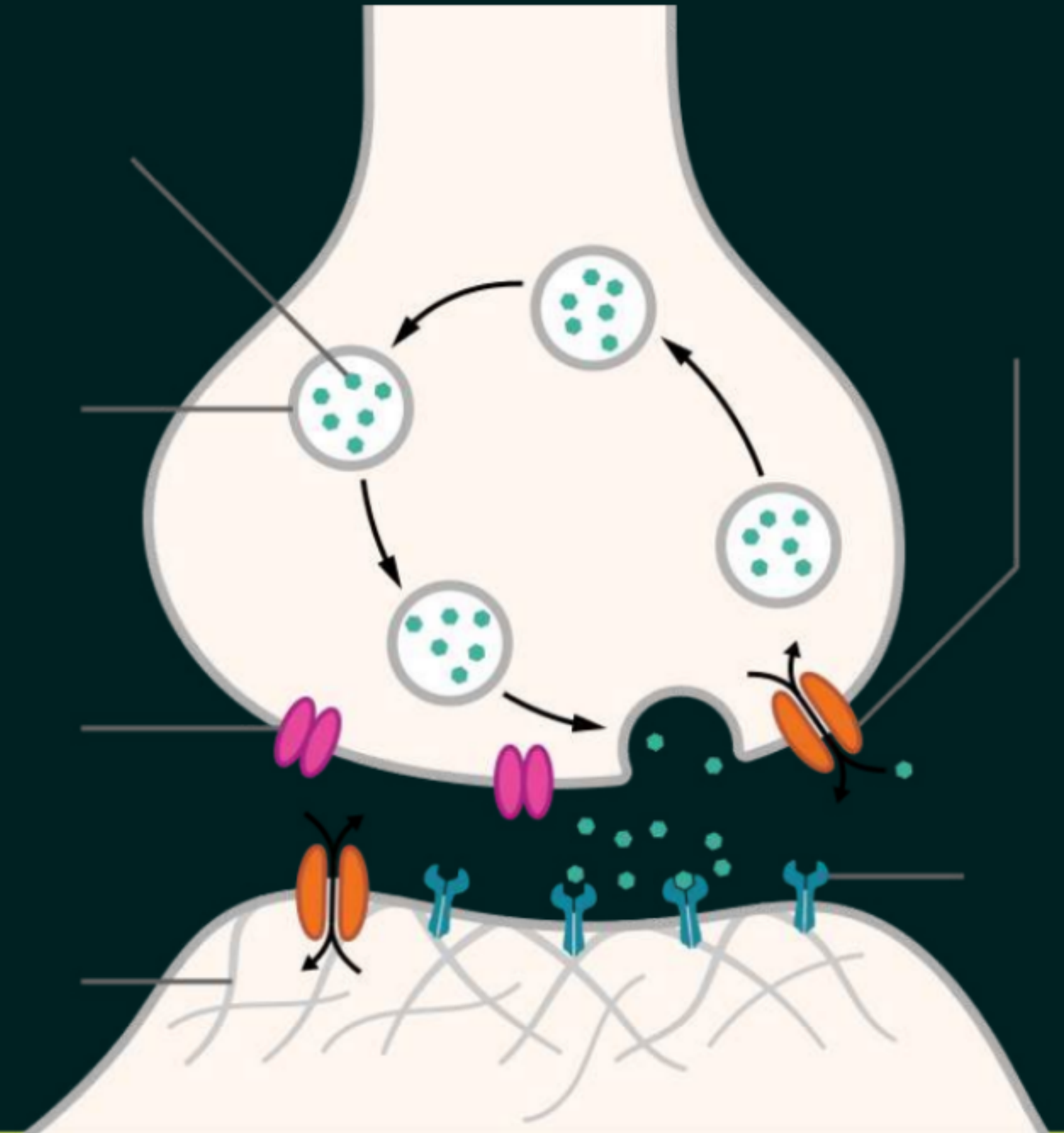
- A neuromuscular junction is the point where the end of a motor neuron connects with a muscle fibre.
- It acts as a synapse that transmits nerve signals to the muscle, causing it to respond or contract.





Neurotransmitters

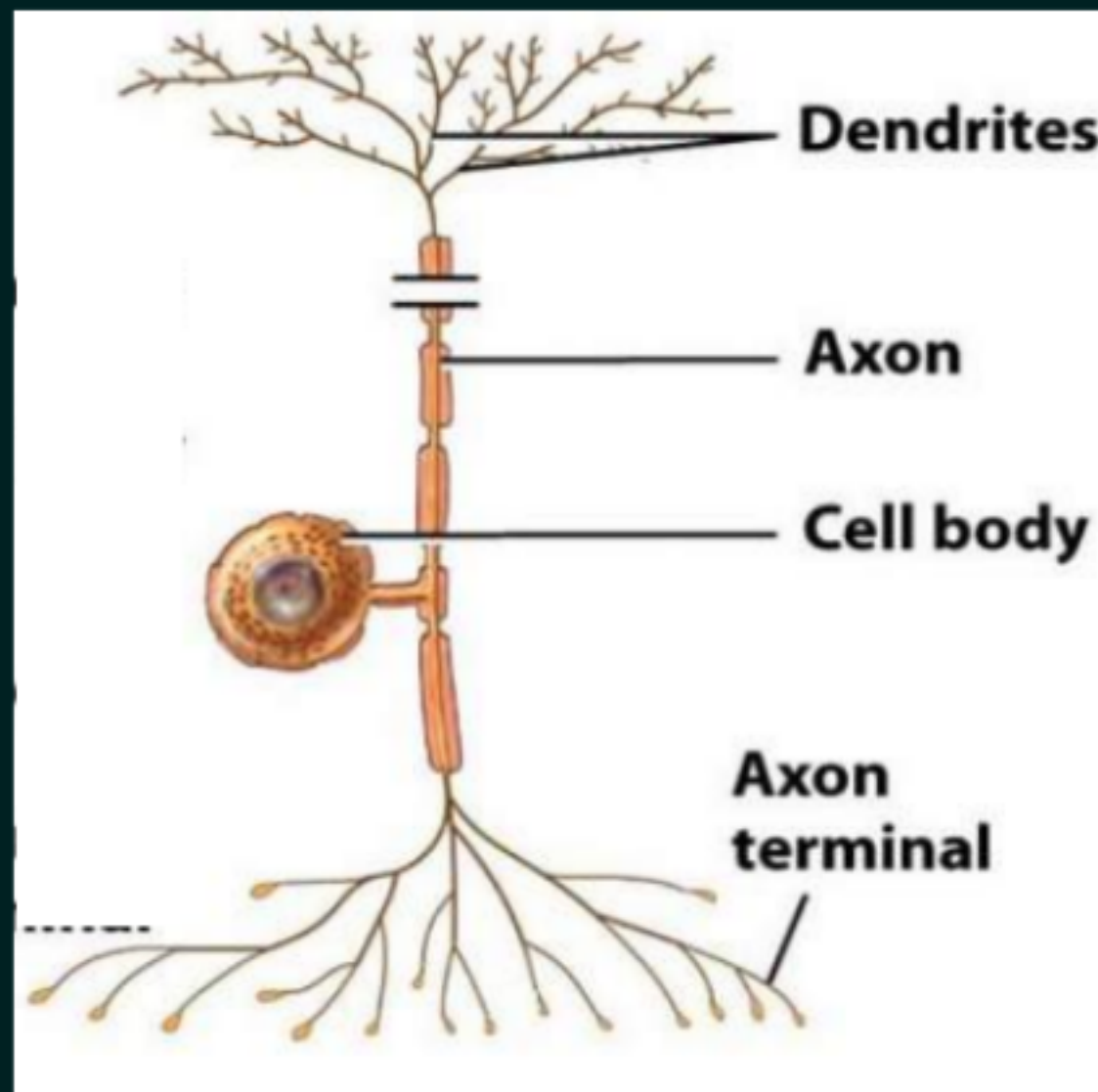
- Neurotransmitters are **chemical messengers** released from the axon terminals of a neuron.
- They help in transmitting the nerve impulse across the **synapse** (gap between two neurons).
- At the synapse, the **electrical impulse is converted into a chemical signal** (neurotransmitter).
- These chemicals quickly cross the gap and generate a new electrical impulse in the next neuron, muscle cell, or gland cell.



Types of Neurons

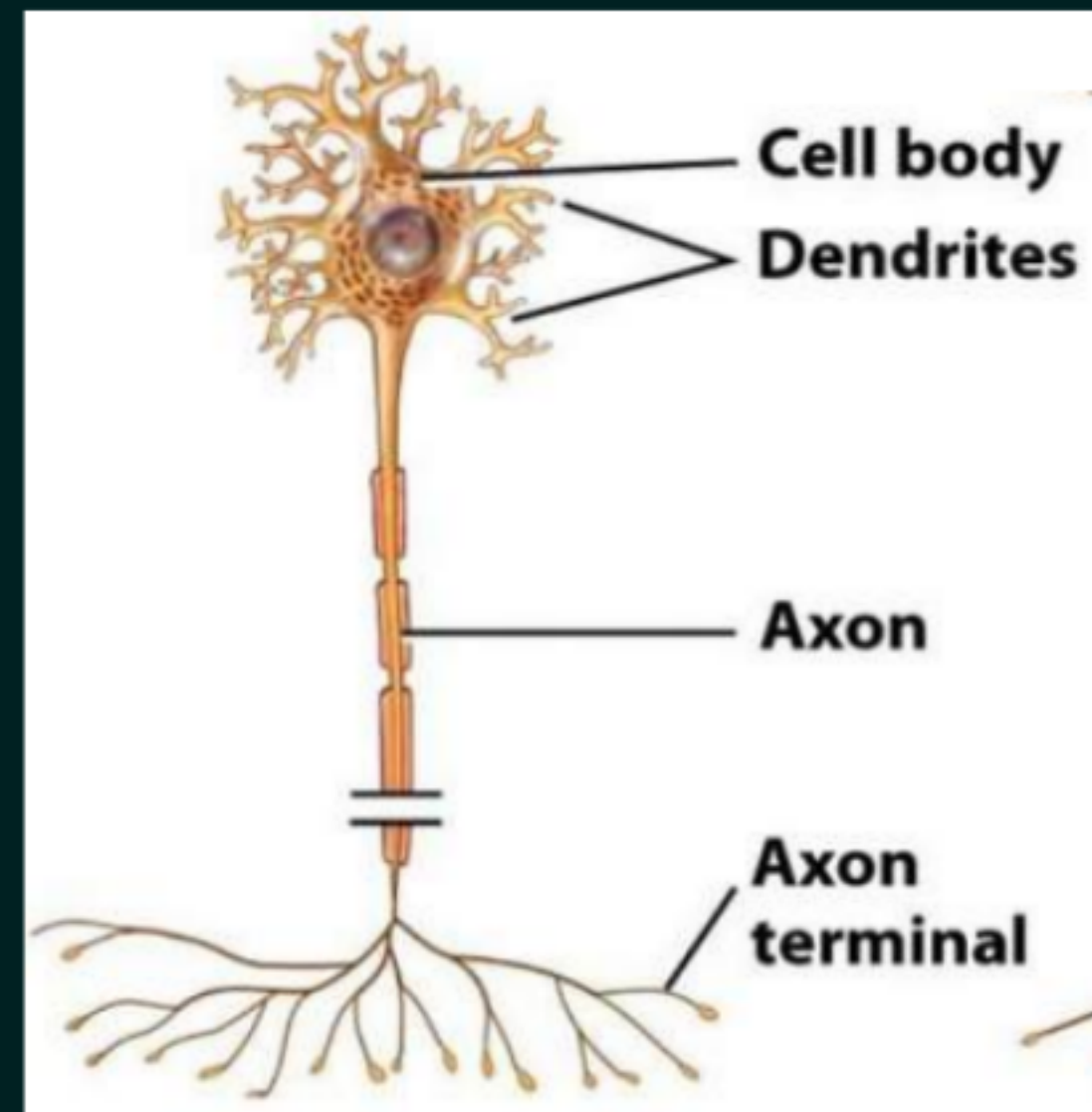
Sensory neurons

Carry messages from sense organs to the brain or spinal cord.



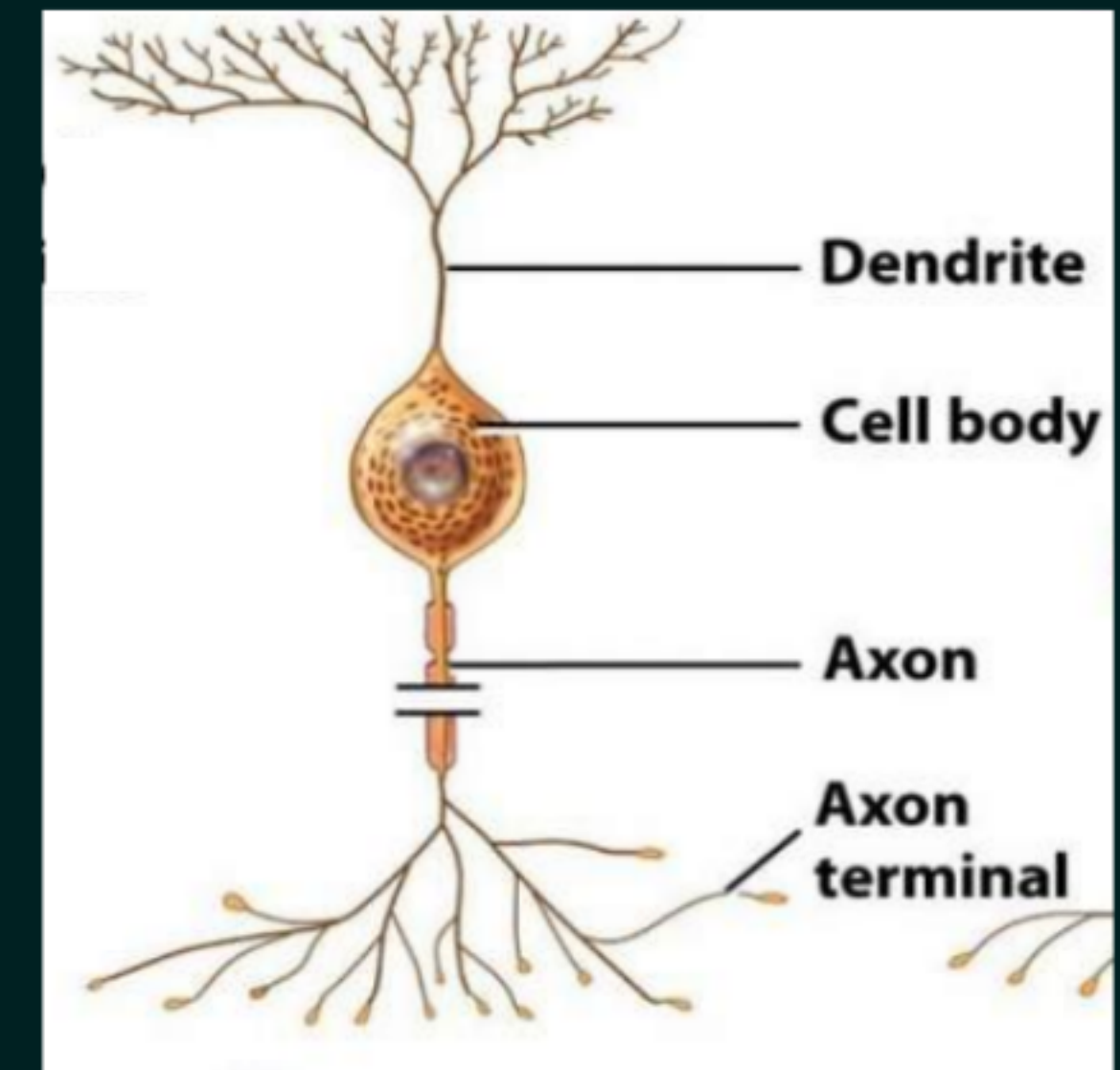
Motor neurons

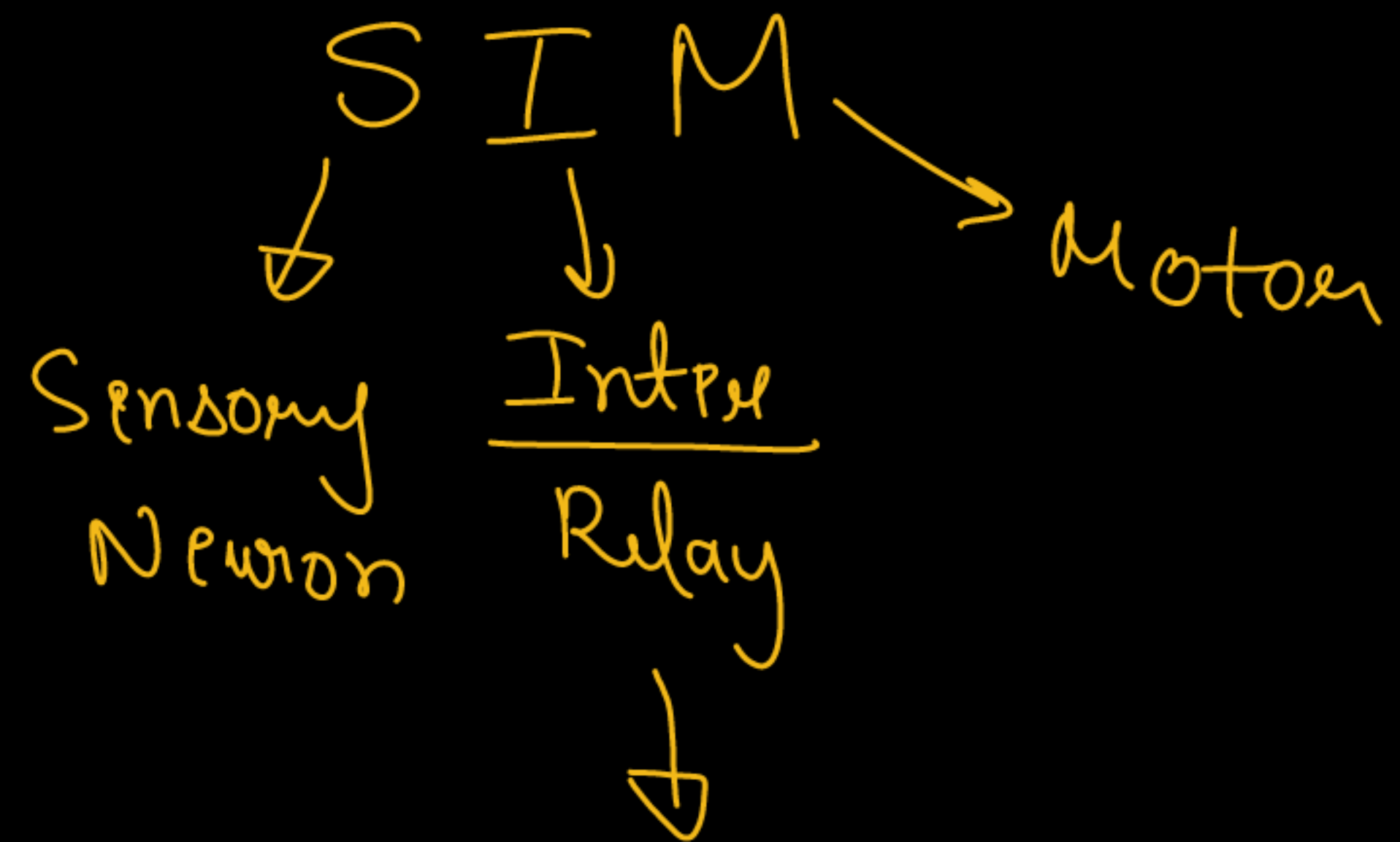
Carry messages from the brain or spinal cord to muscles or glands.



Interneurons

Connect sensory and motor neurons within the brain and spinal cord.





LET'S TEST YOUR MEMORY



Q. What kind of receptors are present in the tongue? (CBSE 2023)



LET'S TEST YOUR MEMORY



①

Q. Assertion (A): In our actions of writing or talking, our nervous system communicates with the muscles.

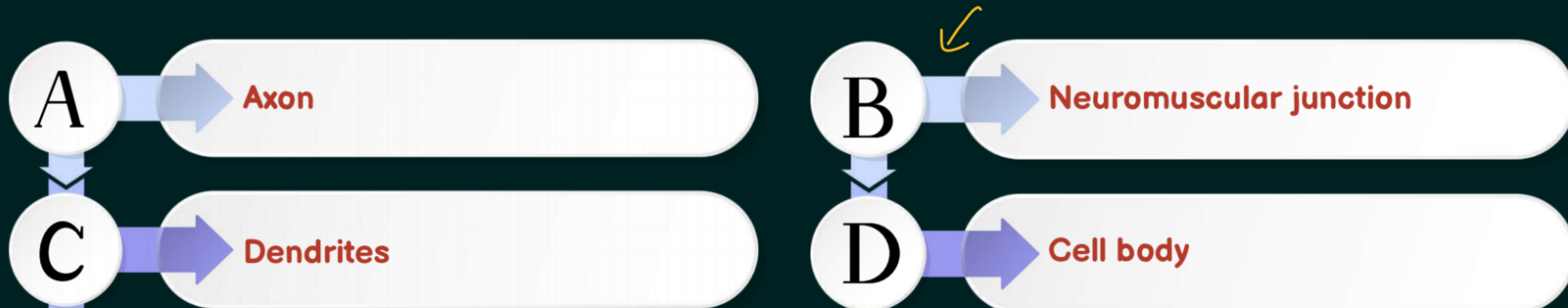
Reason (R): Cranial nerves and spinal nerves form the peripheral nervous system. [CBSE 2025]

- (a) Both A and R are true, and R is the correct explanation of A.
- ☒ (b) Both A and R are true, but R is not the correct explanation of A.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

LET'S TEST YOUR MEMORY



Q. In a nerve cell, the site where the electrical impulse is converted into a chemical signal is known as: [CBSE 2024]



LET'S TEST YOUR MEMORY



Q. Which structure carries impulses away from the cell body?

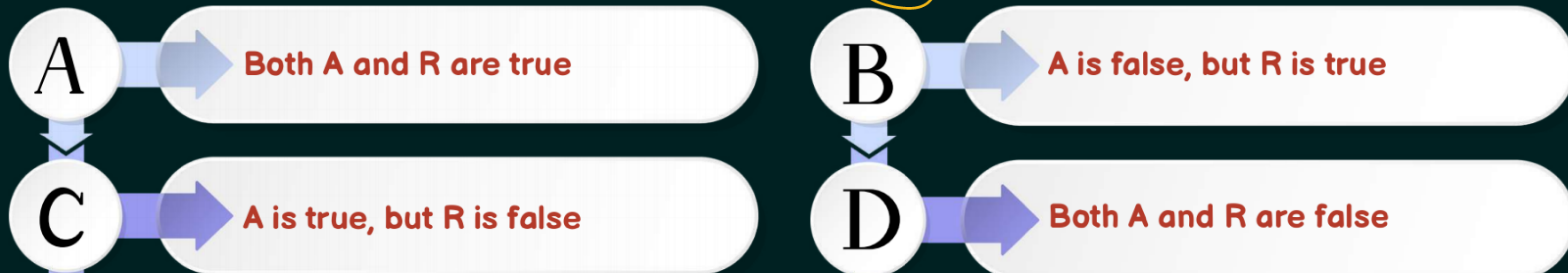


LET'S TEST YOUR MEMORY



~~Assertion (A):~~ Neurons are the shortest cells in the human body.

~~Reason (R):~~ Neurons transmit information throughout the body.



LET'S TEST YOUR MEMORY



Q. (A) Identify the correct statement(s):

- ✓ (I) Taste of sugar is sweet.
- ✓ (II) Gustatory receptors in our tongue are responsible for taste.
- ~~(III) Olfactory receptors in our tongue are responsible for smell.~~
- ~~(IV) Receptors communicate information to the mouth so that we are able to taste things.~~

Options:

- (a) Only (I)
- ✓ (b) Both (I) and (II)
- (c) (I), (II) and (III)
- (d) (I), (III) and (IV)

LET'S TEST YOUR MEMORY



Q. Nervous tissue consists of a well-organised network of nerve cells, known as neurons. These neurons are specialised to transmit information through electrical impulses from one part of the body to another.

Identify the correct neuron parts corresponding to the following functions:

- (I) The part through which information travels as an electrical impulse. → Axon → Nerve ending
- (II) The part where the impulse is converted into a chemical signal for further transmission.
- (III) The part containing the nucleus, major branching fibres, and numerous smaller branching fibres.
- (IV) The part responsible for acquiring information. → D → cell Body

Option	(I)	(II)	(III)	(IV)
(a)	Dendrite	Nerve ending	Axon	Cell body
(b)	Axon	Dendrite	Cell body	Nerve ending
☒ (c)	Axon	Nerve ending	Cell body	Dendrite
(d)	Dendrite	Axon	Nerve ending	Cell body

LET'S TEST YOUR MEMORY



Q. Electrical impulse travels in a neuron from: [CBSE 2025]

- (a) Nerve ending → Axon → Cell body → Dendrite
- ☒ (b) Dendrite → Cell body → Axon → Nerve ending
- (c) Cell body → Dendrite → Axon → Nerve ending
- (d) Dendrite → Axon → Nerve ending → Cell body

LET'S TEST YOUR MEMORY



Q. Which of the following statements about transmission of nerve impulse is INCORRECT?

(a) Nerve impulse travels from the dendritic end towards the axonal end.

~~(b)~~ At the dendritic end, electrical impulses bring about the release of some chemicals, which generate an electrical impulse at the axonal end of another neuron.

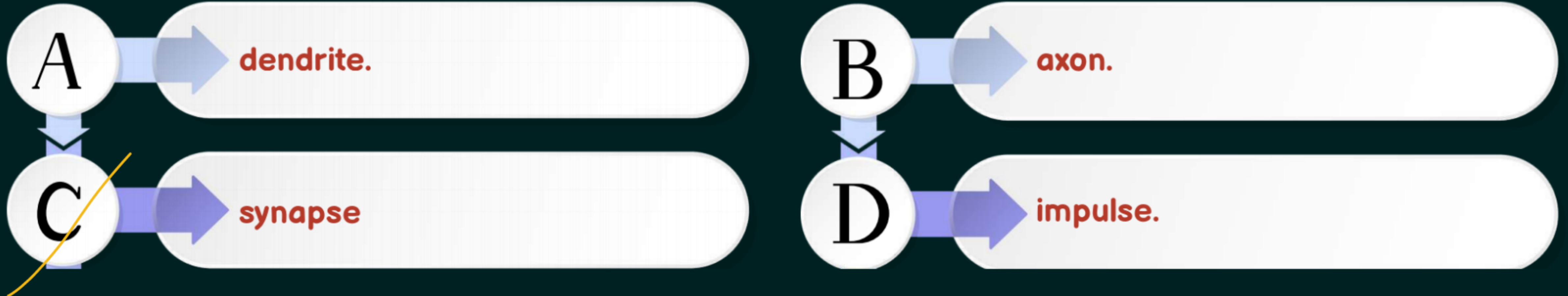
~~(c)~~ The chemicals released from the axonal end of one neuron cross the synapse and generate a similar electrical impulse in a dendrite of another neuron.

(d) A neuron transmits electrical impulses not only to another neuron, but also to muscle and gland cells.

LET'S TEST YOUR MEMORY



Q. The gap between two neurons is called a



LET'S TEST YOUR MEMORY



Q. Draw a diagram of neuron and name and label the part

- (a) where information is acquired,**
- (b) through which information travels as an electric impulse, and**
- (c) where the electric impulse must be converted into a chemical signal for onward transmission.**

(2019 CBSE)

LET'S TEST YOUR MEMORY



Q. What is the function of receptors in our body? Think of situations where receptors do not work properly. What problems are likely to arise?

- Receptors are specialized cells or sense organs that detect stimuli from the surroundings and send information to the nervous system.
- They help the body respond appropriately to changes in the environment.

Problems if receptors do not work properly:

- If photoreceptors in the eyes are damaged, a person may not be able to see properly or may become blind.
- If olfactory receptors in the nose do not function, a person may lose the sense of smell.
- If receptors in the skin are damaged, a person may not feel touch, pain, heat, or cold properly.

LET'S TEST YOUR MEMORY



Q. What happens at the synapse between two neurons?

- At the synapse, the electrical impulse reaching the end of one neuron triggers the release of chemicals called neurotransmitters.
- These chemicals cross the synaptic gap and transmit the signal to the next neuron.
- The next neuron then generates a new electrical impulse.

LET'S TEST YOUR MEMORY



Q. What are the limitations of electrical impulses in multicellular organisms?

- They can reach only those cells that are connected by nervous tissue.
- Their effects are short-lived, so they cannot control slow and continuous processes such as growth and development.

Types of Nervous actions

Type of Nervous Action	Meaning	Control Centre	Speed	Example
Voluntary Action	Actions performed under conscious control	Brain	Usually slower	Writing, walking
Involuntary Action	Actions that occur automatically without conscious control	Brain/Spinal cord	Automatic	Heartbeat, breathing
Reflex Action	Quick automatic response to a stimulus	Mainly spinal cord	Very fast	Pulling hand away from a hot object

Reflex action

Reflex action

- A reflex action is an automatic, sudden, and involuntary response to a stimulus from the environment, occurring without any conscious thought or control.
- Reflex actions help protect the body from harmful situations.
- They are mainly controlled by the spinal cord.
- A reflex action takes place through a **reflex arc** connecting sensory and motor neurons.

Examples:

- I pulled my hand back from the flame reflexly
- I was so hungry my mouth started watering reflexly



Process of reflex action

Stimulus detected by sensory receptors (e.g. touching a hot object)



Sensory neurons carry impulses to the spinal cord



Relay neurons (interneurons) in the spinal cord transfer the signal



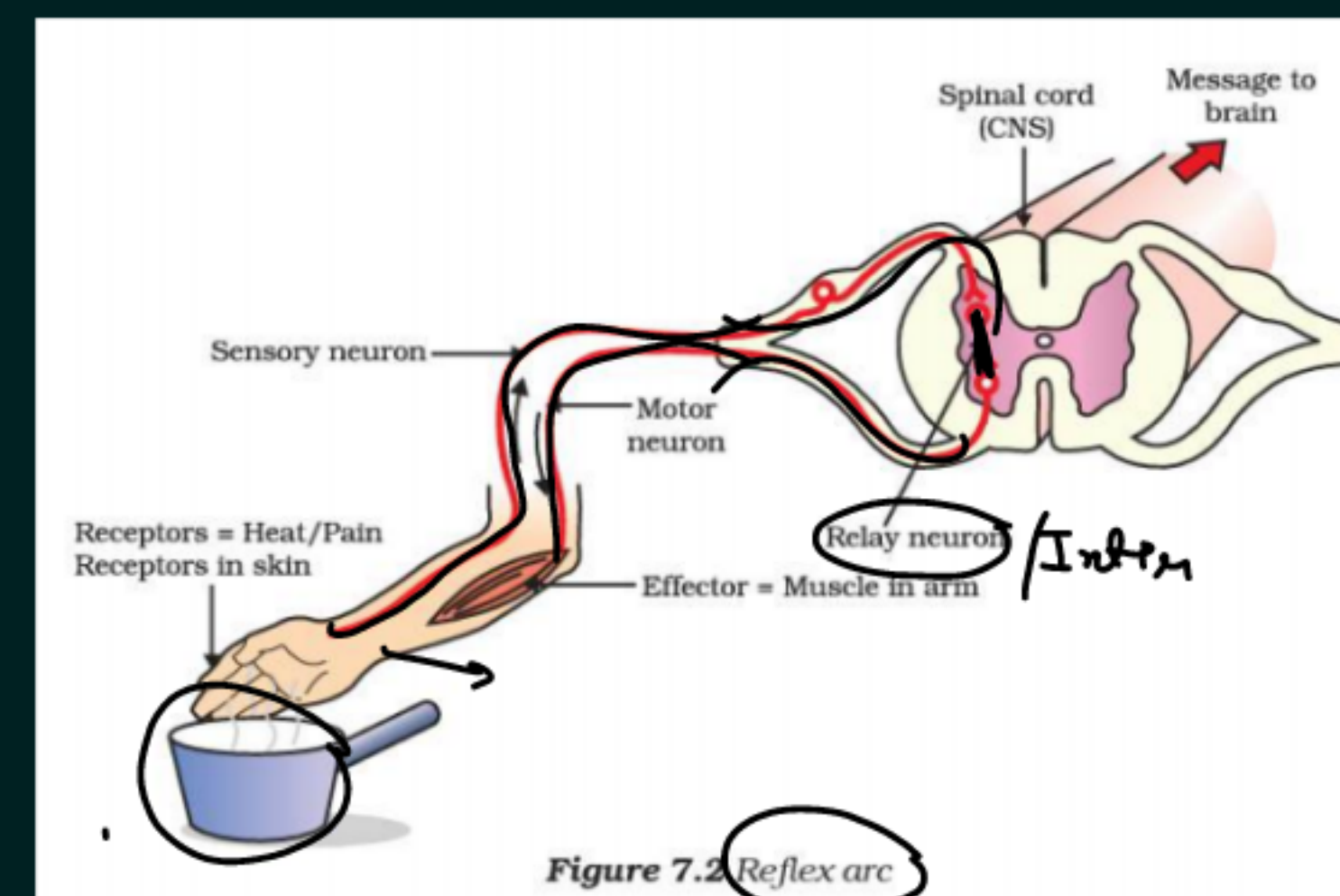
Motor neurons carry impulses to the effector organ



Effector (muscle/gland) responds immediately



Information later reaches the brain for awareness



Role of brain in reflex action



- The brain does not initiate reflex actions.
- Reflex actions are mainly controlled by the spinal cord for a rapid response.
- The brain receives the information after the response occurs.
- It helps in becoming aware of the stimulus and may control future actions.

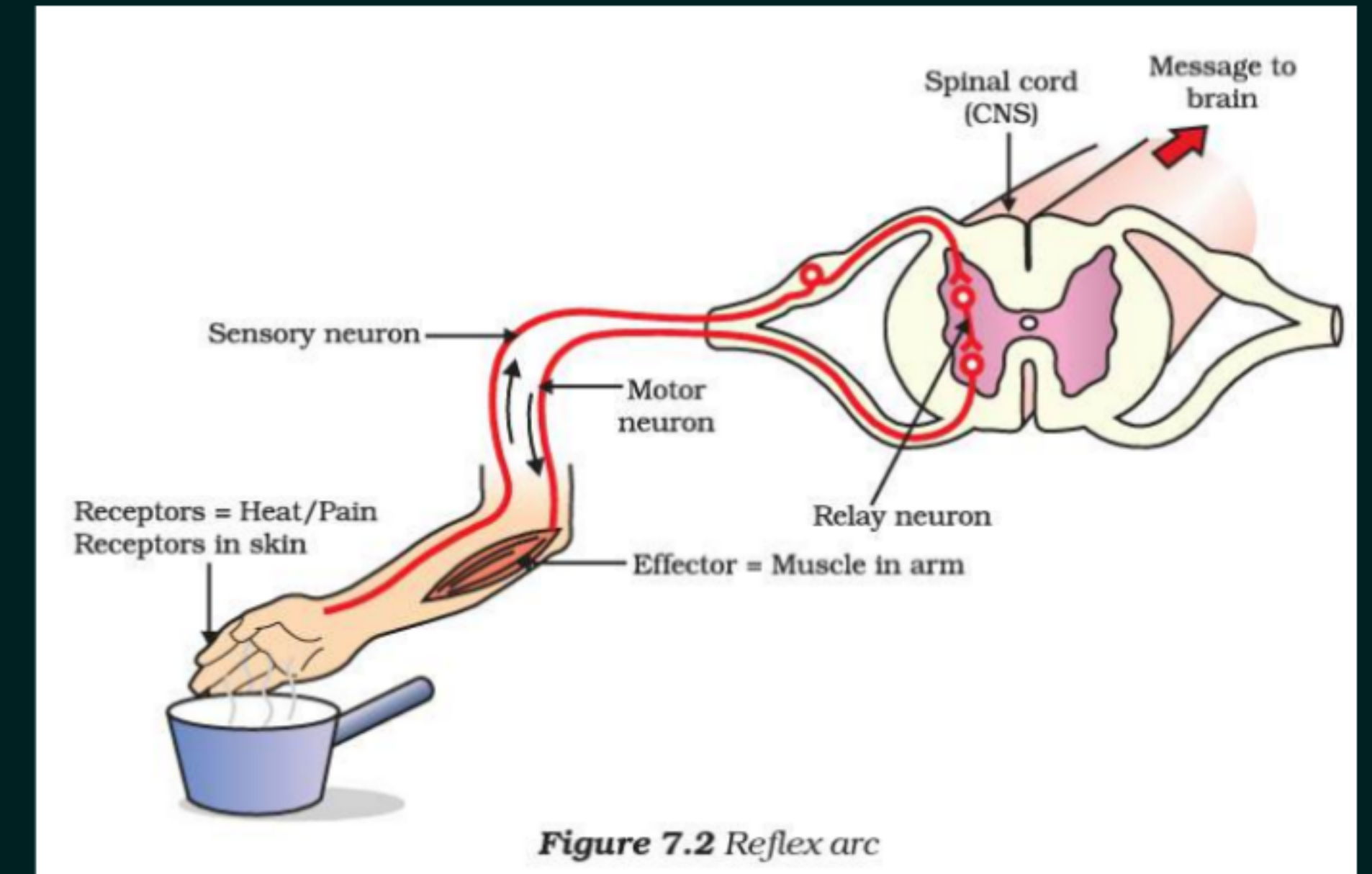


Reflex arc

Reflex arc : Pathway for reflex action connecting sensory nerve to motor nerve.

It includes five main parts:

- ✓ 1. **Stimulus** – e.g. heat detected by skin receptors.
- ✓ 2. **Sensory neuron** – carries message to spinal cord.
- ✓ 3. **Spinal cord** – processes the message.
- ✓ 4. **Motor neuron** – carries command from spinal cord.
- ✓ 5. **Effector (muscle)** – contracts and withdraws the hand.



Importance of Reflex Action

- Provides immediate and automatic response to harmful stimuli.
- Protects the body from injury (e.g., withdrawing hand from hot object).
- Saves time by avoiding delay of thinking in the brain.
- Helps in survival and safety by giving quick reactions in emergencies.



Basis	Reflex Action	Involuntary Action
Meaning	A sudden, quick and automatic response to a stimulus, performed without conscious thinking.	An action that takes place automatically and is not under our conscious control.
Purpose	✓ <u>Mainly protects the body from sudden or dangerous situations.</u>	<u>Maintains the normal internal functioning of the body.</u>
<u>Controlling centre</u>	Usually coordinated through a <u>reflex arc in the spinal cord</u> . Information also reaches the brain.	Mostly controlled by the <u>midbrain and hindbrain</u> . Actions such as blood pressure and vomiting are controlled by the medulla.
Pathway	Stimulus → Receptor → Sensory neuron → Spinal cord → Motor neuron → Effector	Controlled automatically by different regions of the brain and the nerves connected to the concerned organs.
Nature	<u>Generally immediate and short-term.</u>	<u>May continue regularly or rhythmically for a long period.</u>
Examples	Pulling the hand away from a hot object, blinking in bright light and change in pupil size.	Heartbeat, breathing, digestion, blood pressure and vomiting.

LET'S TEST YOUR MEMORY



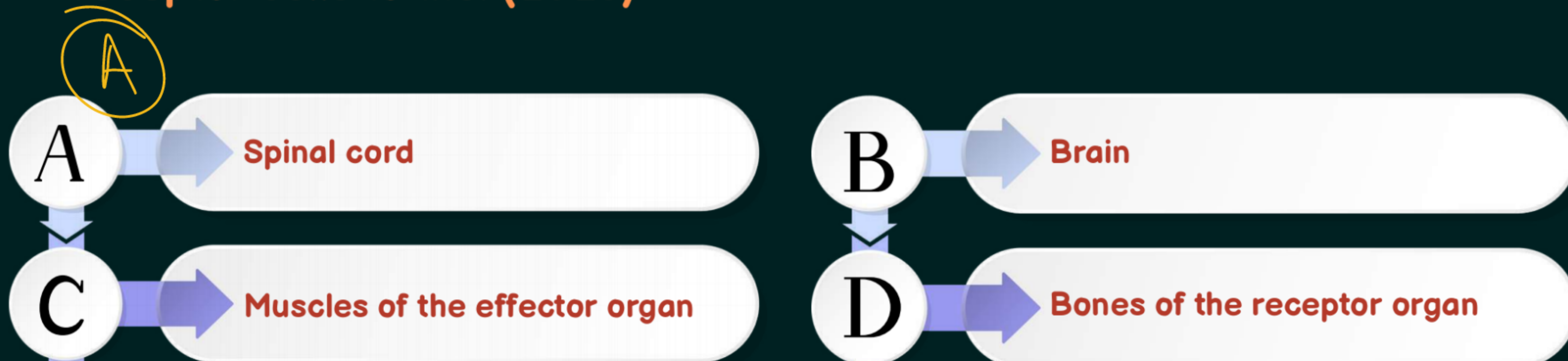
Q. The correct sequence of events when someone's hand touches a hot object unconsciously is: (2024)

- A. Receptors in skin → Motor neuron → Relay neuron → Sensory neuron → Effector muscle in arm
- B. Receptors in skin → Relay neuron → Sensory neuron → Motor neuron → Effector muscle in arm
- ☒ C. Receptors in skin → Sensory neuron → Relay neuron → Motor neuron → Effector muscle in arm
- D. Receptors in skin → Sensory neuron → Effector muscle in arm → Motor neuron → Relay neuron

LET'S TEST YOUR MEMORY



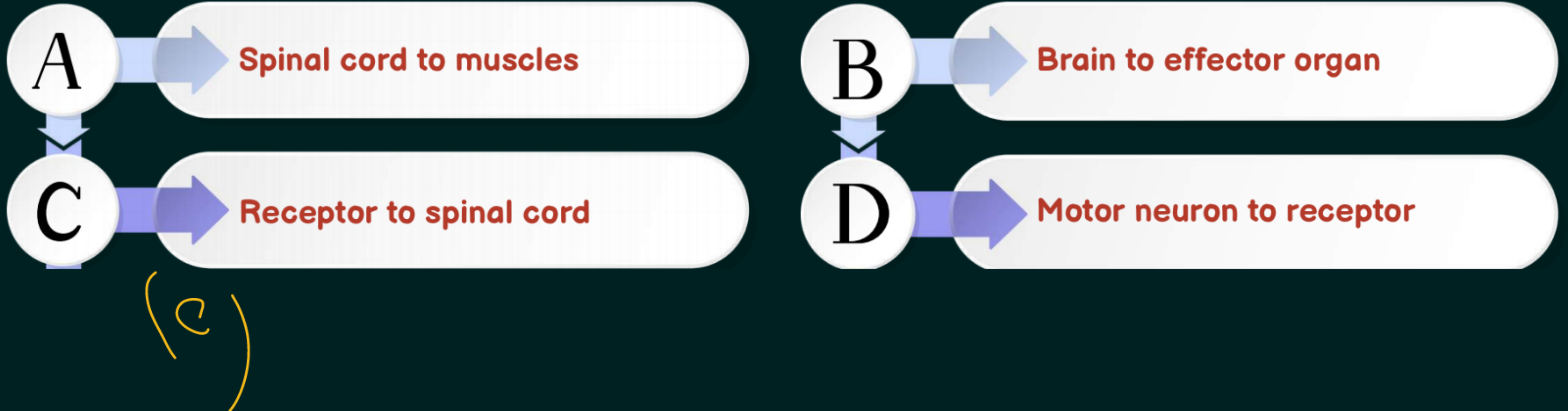
Q. The sensory nerve of a reflex arc carries information from the receptor cells to the: (2023)



LET'S TEST YOUR MEMORY



Q. In a reflex arc, the sensory neuron carries impulses from:



LET'S TEST YOUR MEMORY



In life there are certain changes in the environment called 'stimuli' to which we respond appropriately. Touching a flame suddenly is a dangerous situation for us. One way is to think consciously about the possibility of burning and then moving the hand. But our body has been designed in such a way that we save ourselves from such situations immediately. [CBSE 2025]

Q. (i) Name the action by which we protect ourselves in the situation mentioned above and define it.

(ii) Write the role of (a) motor and (b) relay neuron.

Ans:

(i) Action: Reflex action

Definition: Reflex action is an automatic, immediate, and involuntary response to a stimulus without the involvement of the conscious part of the brain.

Example: Withdrawing the hand instantly on touching a hot flame.

(ii) Role of neurons:

- (a) Motor neuron: Carries impulses from the spinal cord to the effector organs (muscles or glands) to produce a response.
- (b) Relay neuron (interneuron): Connects the sensory neuron to the motor neuron in the spinal cord and helps in transmission of impulses



LET'S TEST YOUR MEMORY

Q: (a) What is meant by the reflex arc? Where are they formed in the human body? (b) Why have reflex arcs evolved in animals? [CBSE 2025]

ANS:

Reflex arc: The pathway through which nerve impulses travel during a reflex action

✓ It consists of: Receptor → Sensory neuron → Spinal cord → Motor neuron → Effector (muscle/gland)

- Reflex arcs are formed in the spinal cord of the human body.

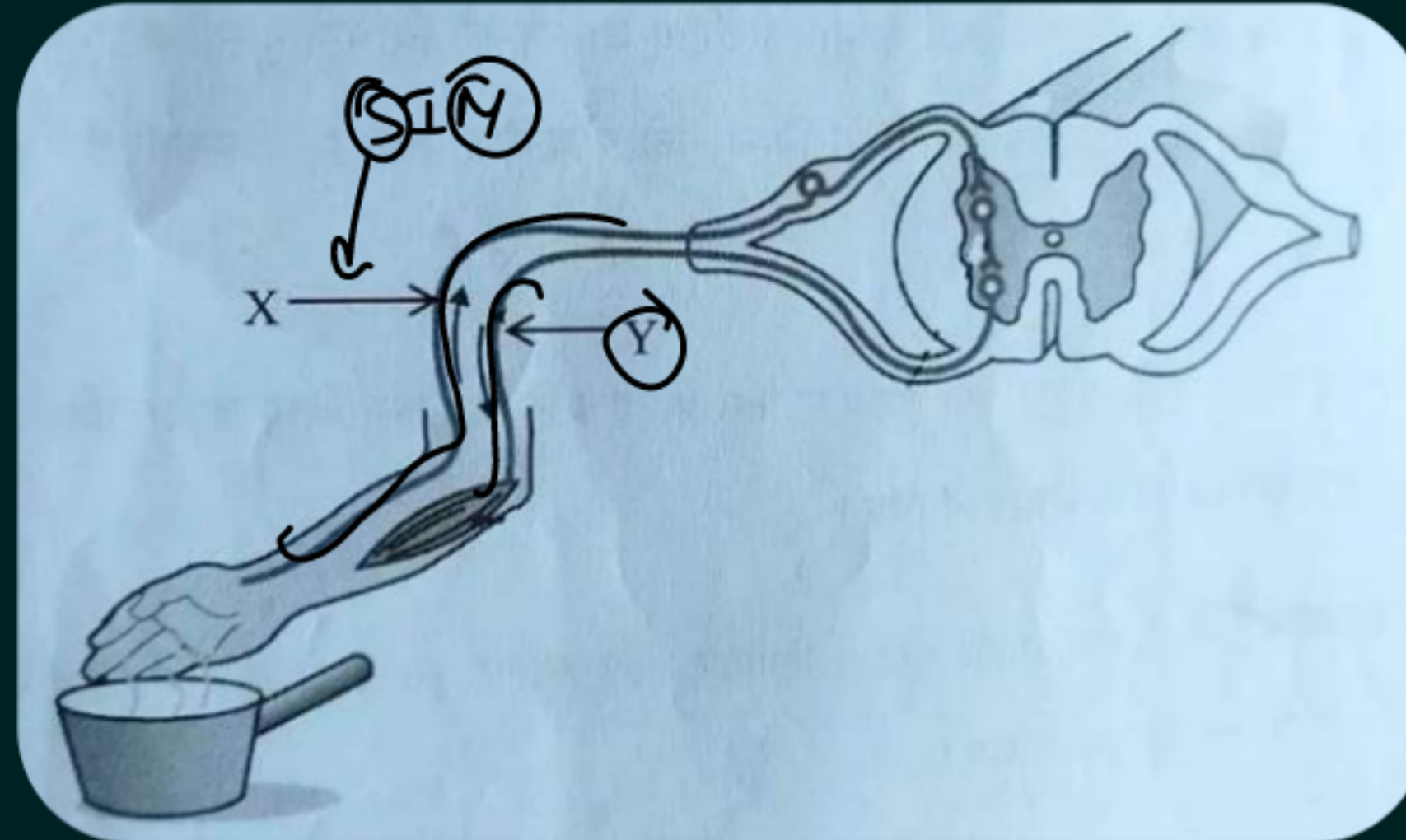
Reflex arcs have evolved in animals:

- Reflex arcs enable quick, automatic, and involuntary responses to harmful stimuli.
- They protect the body from injury without waiting for instructions from the brain.
- Example: Immediate withdrawal of the hand from a hot object.

LET'S TEST YOUR MEMORY



Q. In the given diagram, name the labelled parts X and Y. Mention one function of each.



LET'S TEST YOUR MEMORY



HW

Q. How have reflex arcs evolved in animals? Why are they important?

Ans. Reflex arcs have evolved in animals to provide rapid and automatic responses to harmful or sudden stimuli. This helps animals to protect themselves quickly without waiting for the brain to process the information.

- Located mainly in the spinal cord, reflex arcs skip the brain's detailed processing.
- This makes the reaction faster and more efficient.
- They give animals a survival advantage by helping them escape danger or prevent injury in emergency situations.

☆ PYQ

आरंभ
BATCH

LET'S TEST YOUR MEMORY



Q. Write the sequence of events that involve the response of a person when a dust particles is inhaled through the nose by him

When dust particles are inhaled through the nose, the following sequence of events occurs:

1. Dust particles irritate the lining of the nasal cavity.
2. Sensory receptors in the nose get stimulated.
3. The sensory nerve carries the impulse to the brain.
4. The brain sends motor impulses to the muscles involved in sneezing.
5. Sneezing occurs, expelling the dust particles from the nose.

LET'S TEST YOUR MEMORY



Q. A student accidentally touches a hot utensil and immediately withdraws her hand.

(a) Name the type of action involved.

(b) Why is this response completed before the person consciously feels pain?

(c) Name the centre that coordinates this response.

(d) Trace the complete pathway followed by the nerve impulse.

(e) What role does the brain play in this situation?

a) It is a reflex action—a quick, automatic and involuntary response to a stimulus.

(b) The response is completed before conscious pain is felt because the impulse is processed immediately by the spinal cord, without waiting for the brain to analyse it.

(c) The spinal cord coordinates this reflex action.

(d) Pathway of the nerve impulse:

Hot utensil → Heat/pain receptors in skin → Sensory neuron → Relay neuron in spinal cord → Motor neuron → Muscles of the arm → Hand is withdrawn

This pathway is called the reflex arc.

(e) At the same time, an impulse is also sent from the spinal cord to the brain. The brain interprets the stimulus as pain and makes the person consciously aware of what has happened. The brain is informed after the immediate protective response has already begun.

***Kal
Milte h!***

